

Essential Spectral Accumulation and Optimal Damping for a Kelvin–Voigt Wave

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Abstract

We analyze the Dirichlet Kelvin–Voigt equation on the normalized interval $(0, \pi)$ at its physical time clock. Sine modes give an exact quadratic pencil. As the damping crosses $bn = 2$, modes pass through underdamped, critical-Jordan, and overdamped regimes. The overdamped slow roots approach the non-eigenvalue essential spectral point $-1/b$, while the fast roots escape to minus infinity. Comparing this high-frequency mechanism with the first-mode real part yields the exact spectral-abscissa gap $\min(b/2, 1/b)$ and its unique optimizer $b = \sqrt{2}$. We prove the energy identity and explicitly distinguish spectral abscissa from a uniform operator-norm decay assertion at a Jordan parameter. Independent numerical, symbolic, replay, and hostile-mutation controls accompany the theorem.

1 Model and generator

Let

$$u_{tt} - u_{xx} - b u_{txx} = 0, \quad u(0, t) = u(\pi, t) = 0, \quad b \geq 0.$$

The energy is

$$E(t) = \frac{1}{2} \int_0^\pi (|u_t|^2 + |u_x|^2) dx.$$

For $b > 0$, the first-order generator on $H_0^1(0, \pi) \times L^2(0, \pi)$ is $A(u, v) = (v, (u + bv)_{xx})$ with exact domain

$$D(A) = \{(u, v) : v \in H_0^1(0, \pi), u + bv \in H^2(0, \pi) \cap H_0^1(0, \pi)\}.$$

This domain statement is kept separate from the spectral-abscissa calculation.

2 Root atlas

Writing $u(x, t) = \sum_{n \geq 1} q_n(t) \sin(nx)$ gives

$$q_n'' + bn^2 q_n' + n^2 q_n = 0, \quad \lambda_{n, \pm} = \frac{-bn^2 \pm \sqrt{b^2 n^4 - 4n^2}}{2}.$$

Theorem 1 (Kelvin–Voigt root, essential point, and gap). *For $b > 0$, $bn < 2$ is underdamped, $bn = 2$ is a defective Jordan root $-n$, and $bn > 2$ is overdamped. There are finitely many underdamped modes and at most one critical mode. In the overdamped regime,*

$$\lambda_{n, +} = -\frac{2n^2}{bn^2 + \sqrt{b^2 n^4 - 4n^2}} \longrightarrow -\frac{1}{b} \text{ from below,} \quad \lambda_{n, -} \longrightarrow -\infty.$$

We use the Weyl singular-sequence definition of essential spectrum. The point $-1/b$ belongs to it as an accumulation point but is not an eigenvalue of the energy generator. Indeed, substituting $\lambda = -1/b$ into

$$(\lambda^2 - (1 + b\lambda)\partial_{xx})u = 0$$

leaves $b^{-2}u = 0$, so a Dirichlet eigenfunction must vanish. The normalized high-mode sine eigenvectors form a singular sequence (weakly converging to zero with generator residual tending to zero), and the positive-time semigroup is not compact or Schatten.

The spectral-abscissa gap is exactly

$$\gamma(b) = -\sup\{\Re\lambda_{n,\pm} : n \geq 1\} = \min\left(\frac{b}{2}, \frac{1}{b}\right).$$

It has the unique maximizer $b_\star = \sqrt{2}$ and $\gamma(b_\star) = 1/\sqrt{2}$. This is a spectral-abscissa statement; at a critical Jordan parameter it does not assert a uniform operator-norm bound with the exact exponent. Finally,

$$E'(t) = -b \int_0^\pi |u_{tx}|^2 dx \leq 0.$$

At $b = 0$, $\lambda_{n,\pm} = \pm in$ and the undamped wave group is unitary.

Proof. The sine expansion proves the quadratic pencil. The discriminant gives the three regimes and the double-root Jordan form. For an overdamped mode, rationalizing the plus root gives the displayed expression; it is strictly below $-1/b$, and its limit is $-1/b$. In the underdamped range the largest real part is $-bn^2/2$, maximized at $n = 1$; in the overdamped range every slow root is below $-1/b$. Therefore the supremum of real parts is $\max(-b/2, -1/b)$, proving the gap formula. The two positive branches $b/2$ and $1/b$ meet only at $b = \sqrt{2}$, which is the unique maximizer. Substitution at $-1/b$ proves non-eigenvalue status directly, while the normalized high-mode eigenvectors give the singular/noncompact sequence. Multiplication of the PDE by u_t , integration by parts, and the Dirichlet boundary conditions give the energy identity. $\square$

3 Boundary and interpretation

The critical faces $b = 2/n$ are retained rather than smoothed away. A Jordan block produces a polynomial prefactor in its modal solution, so the spectral-abscissa gap should not be silently upgraded to an exact uniform norm-decay constant. The point $-1/b$ is a limit point of eigenvalues, not a hidden eigenmode. The $b = 0$ unitary problem is a separate boundary model.

Source note

Kelvin–Voigt spectral context includes Guo, Wang, and Zhang [1] and Jing Wang and Jun-Min Wang [2]. Damped-string decay and interval determinant context are recorded by Cox and Zuazua [3] and Freitas and Lipovský [4]; none is used to assert an unproved norm estimate.

References

- [1] B. Z. Guo, J. M. Wang, and G. D. Zhang, *Spectral analysis of a wave equation with Kelvin–Voigt damping*, *ZAMM* 90 (2010), 323–342. DOI: 10.1002/zamm.200900275.
- [2] Jing Wang and Jun-Min Wang, *Spectral analysis and exponential stability of one-dimensional wave equation with viscoelastic damping*, *Journal of Mathematical Analysis and Applications* 410 (2014), 499–512. DOI: 10.1016/j.jmaa.2013.08.034.
- [3] S. Cox and E. Zuazua, *The rate at which energy decays in a damped string*, *Communications in Partial Differential Equations* 19 (1994), 213–243. DOI: 10.1080/03605309408821015.
- [4] P. Freitas and J. Lipovský, *Spectral determinant for the damped wave equation on an interval*, *Acta Physica Polonica A* 136 (2019), 817–823. DOI: 10.12693/APhysPolA.136.817.

Declarations

Scope. NO_BAD_EULER_OR_ROOT_NUMBER; no target arithmetic or operator claim. **Data and code.** The theorem, ledger, independent checks, and build instructions are released with HCS-C218. **AI-use disclosure.** Generative tools assisted drafting and code generation; the internal artifact chain checked the displayed claims and metadata. This is not external peer review.